SMARTBRIDGE
Let's Bridge the Gap

Date	28-10-2026
Team ID	**EN2EL3010078**
Project Name	EV Market Performance and Insights
Maximum Marks	2 Marks

Project Design Phase - II

Technology Stack (Architecture & Stack)

Define Your Technology Stack:
Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S. No	Component	Description	Technology	Justification / Purpose
1.	**Programming / Query Language**	Language used to query, filter, transform, aggregate, and analyze data.	**Tableau calculated fields and LOD expressions**	Used to create measures such as CO2 saved and retained value, and to filter, aggregate and compare data by category, manufacturer, battery type and region.
2.	**Database / Data Warehouse**	Warehouse used to store and manage structured business data for analysis and reporting.	**Excel / CSV file (EV market dataset)**	Holds the structured data on sales, price, battery, range, manufacturer, country, ratings and CO2, and connects to Tableau without needing a database server.

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3.	**Data Ingestion**	Used to load data from source files or external systems into the analytical environment.	**Tableau Desktop data source connection (Excel / Text file)**	Loads the dataset straight into Tableau and can be refreshed whenever the source file is updated.
4.	**Data Transformation and Business Logic**	Tools/Logic used to clean, join, filter, aggregate data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	**Tableau Desktop: Data Source tab, calculated fields and aggregations**	Manages data types, relationships and calculations for KPIs such as total sales, average price, average range, retained value and CO2 saved.
5.	**Data Visualization / Analytics**	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	**Tableau Desktop: bar, line, box plot, bubble, scatter, dual-axis and stacked area charts**	Creates the interactive visuals for category sales, manufacturer performance, battery technology, driving range, charging efficiency and regional CO2 impact.
6.	**Front-End / User Interface**	Interface through which users interact with dashboards, filters, parameters, and visual insights.	**Tableau Dashboards (Dashboard 1 and Dashboard 2), shared through Tableau Public**	Gives executives and analysts interactive filters, legend highlighting and tooltips without coding, and lets stakeholders open the dashboards in a browser.